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Batch: 29

Assessment: 18.3

Task 1: Public Transport Route Fare API

Prompt:

Generate Python code to fetch transport fare between two stations using an API with error handling.

Code:

```python
#Generate Python code to fetch transport fare between two stations using an API with error handling.
import requests
def fetch_transport_fare(api_url, origin, destination):
    try:
        response = requests.get(api_url, params={'origin': origin, 'destination': destination}, timeout=5)
        response.raise_for_status()
        data = response.json()
        fare = data.get('fare')

        if fare is not None:
            return fare
        else:
            print("Fare not available from API")
    except requests.exceptions.RequestException as e:
        print("API failed:", e)
        print("Using fallback data...")
    mock_data = {
        ("Station A", "Station B"): 50,
        ("Chennai", "Bangalore"): 500
    }
    return mock_data.get((origin, destination))
# Example usage
api_url = "https://api.transportfare.com/getFare"
origin = "Station A"
destination = "Station B"

fare = fetch_transport_fare(api_url, origin, destination)

if fare is not None:
    print(f"The fare from {origin} to {destination} is: {fare}")
else:
    print("Could not fetch the fare information.")
```

Output:

```
API failed: HTTPSConnectionPool(host='api.transportfare.com', port=443): Max retries exceeded with url: /getFare?origin=Station+A&destination=Station+B (Caused by NameReso
lutionError("HTTPSConnection(host='api.transportfare.com', port=443): Failed to resolve 'api.transportfare.com' ([Errno 11001] getaddrinfo failed)"))
Using fallback data...
lutionError("HTTPSConnection(host='api.transportfare.com', port=443): Failed to resolve 'api.transportfare.com' ([Errno 11001] getaddrinfo failed)"))
Using fallback data...
The fare from Station A to Station B is: 50
```

Significance:

This task demonstrates how to integrate external APIs to fetch real-time data. It highlights handling of API failures and ensures program reliability using fallback data. It is useful in building transport and travel applications.

Task 2: Currency Exchange Rates

Prompt:

Generate Python code to convert INR to USD, EUR, and GBP using an API with input validation and error handling.

Code:

```python
#Generate Python code to convert INR to USD, EUR, and GBP using an API with input validation and error handling.
import requests
def convert_inr_to_foreign_currency(api_url, amount_in_inr):
    if amount_in_inr < 0:
        print("Amount cannot be negative.")
        return None
    try:
        response = requests.get(api_url, params={'amount': amount_in_inr}, timeout=5)
        response.raise_for_status()
        data = response.json()
        usd = data.get('USD')
        eur = data.get('EUR')
        gbp = data.get('GBP')

        if usd is not None and eur is not None and gbp is not None:
            return {'USD': usd, 'EUR': eur, 'GBP': gbp}
        else:
            print("Currency conversion data not available from API")
    except requests.exceptions.RequestException as e:
        print("API failed:", e)
        print("Using fallback conversion rates...")
    mock_conversion_rates = {
        'USD': 0.013,
        'EUR': 0.011,
        'GBP': 0.009
    }
    return {currency: amount_in_inr * rate for currency, rate in mock_conversion_rates.items()}
# Example usage
api_url = "https://api.currencyconverter.com/convert"
amount_in_inr = 1000
conversion_result = convert_inr_to_foreign_currency(api_url, amount_in_inr)
if conversion_result is not None:
    print(f"{amount_in_inr} INR is equivalent to: {conversion_result}")
else:
    print("Could not fetch the conversion information.")
```

Output:

```
API failed: HTTPSConnectionPool(host='api.currencyconverter.com', port=443): Max retries exceeded with url: /convert?amount=1000 (Caused by NameResolutionError("HTTPSConnection(host='api.currency
converter.com', port=443): Failed to resolve 'api.currencyconverter.com' ([Errno 11002] getaddrinfo failed)"))
Using fallback conversion rates...
1000 INR is equivalent to: {'USD': 13.0, 'EUR': 11.0, 'GBP': 9.0}
```

Significance:

This task shows how to retrieve and process real-time financial data from APIs. It helps in understanding input validation and error handling in API responses. It is useful for finance and e-commerce applications.

Task 3: GitHub Repository Info Fetcher

Prompt:

Generate Python code to fetch GitHub repository details (stars, forks, issues) using API with error handling.

Code:

```python
#Generate Python code to fetch GitHub repository details (stars, forks, issues) using API with error handling.
import requests
def fetch_github_repo_details(api_url, owner, repo):
    try:
        response = requests.get(f"{api_url}/repos/{owner}/{repo}", timeout=5)
        response.raise_for_status()
        data = response.json()
        stars = data.get('stargazers_count')
        forks = data.get('forks_count')
        issues = data.get('open_issues_count')

        if stars is not None and forks is not None and issues is not None:
            return {'stars': stars, 'forks': forks, 'issues': issues}
        else:
            print("Repository details not available from API")
    except requests.exceptions.RequestException as e:
        print("API failed:", e)
        print("Using fallback data...")
    mock_data = {
        ("octocat", "Hello-World"): {'stars': 1500, 'forks': 300, 'issues': 50},
        ("torvalds", "linux"): {'stars': 100000, 'forks': 40000, 'issues': 500}
    }
    return mock_data.get((owner, repo))
# Example usage
api_url = "https://api.github.com"
owner = "octocat"
repo = "Hello-World"
repo_details = fetch_github_repo_details(api_url, owner, repo)
if repo_details is not None:
    print(f"Repository '{owner}/{repo}' details: {repo_details}")
else:
    print("Could not fetch the repository details.")
```

Output:

```
Repository 'octocat/Hello-World' details: {'stars': 3550, 'forks': 5948, 'issues': 3797}
```

Significance:

This task demonstrates interaction with developer APIs to fetch repository details. It highlights handling of errors like invalid input and rate limits. It is useful for building developer tools and dashboards.

Task 4: Real-Time Application: News Headlines Aggregator

Prompt:

Generate Python code to fetch top 5 news headlines by category using API with retry and error handling.

Code:

```python
#Generate Python code to fetch top 5 news headlines by category using API with retry and error handling.
import requests
def fetch_top_news_headlines(api_url, category):
    try:
        response = requests.get(f"{api_url}/top-headlines", params={'category': category}, timeout=5)
        response.raise_for_status()
        data = response.json()
        headlines = [article['title'] for article in data.get('articles', [])][:5]

        if headlines:
            return headlines
        else:
            print("No headlines available from API")
    except requests.exceptions.RequestException as e:
        print("API failed:", e)
        print("Using fallback data...")
    mock_data = {
        "technology": ["Tech News 1", "Tech News 2", "Tech News 3", "Tech News 4", "Tech News 5"],
        "sports": ["Sports News 1", "Sports News 2", "Sports News 3", "Sports News 4", "Sports News 5"]
    }
    return mock_data.get(category)
# Example usage
api_url = "https://newsapi.org/v2"
category = "technology"
headlines = fetch_top_news_headlines(api_url, category)
if headlines is not None:
    print(f"Top 5 news headlines in '{category}': {headlines}")
else:
    print("Could not fetch the news headlines.")
```

Output:

```
API failed: 401 Client Error: Unauthorized for url: https://newsapi.org/v2/top-headlines?category=technology
Using fallback data...
Top 5 news headlines in 'technology': ['Tech News 1', 'Tech News 2', 'Tech News 3', 'Tech News 4', 'Tech News 5']
```

Significance:

This task shows how to fetch and display dynamic content from news APIs. It includes retry mechanisms and error handling for reliable data fetching. It is useful for building real-time news and content applications.

Task 5: COVID-19 Statistics API Integration

Prompt:

Generate Python code to fetch COVID-19 statistics by country using API with retry and error handling.

Code:

```python
#Generate Python code to fetch COVID-19 statistics by country using API with retry and error handling.
import requests
def fetch_covid_stats(api_url, country):
    try:
        response = requests.get(f"{api_url}/countries/{country}", timeout=5)
        response.raise_for_status()
        data = response.json()
        stats = {
            'cases': data.get('cases'),
            'deaths': data.get('deaths'),
            'recovered': data.get('recovered')
        }

        if all(value is not None for value in stats.values()):
            return stats
        else:
            print("COVID-19 statistics not available from API")
    except requests.exceptions.RequestException as e:
        print("API failed:", e)
        print("Using fallback data...")
    mock_data = {
        "India": {'cases': 30000000, 'deaths': 400000, 'recovered': 29000000},
        "USA": {'cases': 35000000, 'deaths': 600000, 'recovered': 34000000}
    }
    return mock_data.get(country)
# Example usage
api_url = "https://disease.sh/v3/covid-19"
country = "India"
covid_stats = fetch_covid_stats(api_url, country)
if covid_stats is not None:
    print(f"COVID-19 statistics for '{country}': {covid_stats}")
else:
    print("Could not fetch the COVID-19 statistics.")
```

Output:

```
COVID-19 statistics for 'India': {'cases': 45035393, 'deaths': 533570, 'recovered': 0}
```

Significance:

This task demonstrates fetching real-time health data from public APIs. It highlights handling of rate limits and network errors effectively. It is useful for health monitoring and data analysis applications.